

Statistical Data Analysis

Organisation

Marc Corstanje

Course Structure

Lectures & Practicals

- › The course has two main purposes:
 1. acquire the knowledge contained in the extensive [Syllabus](#), which contains all essential theory and also many practical illustrations
 2. apply the acquired theoretical knowledge to practical problems using  (including writing reports about data analyses)
- › [Lecture](#): on Wednesday from 15:30 to 17:15 in Theater3-NU-2C33
- › [Practical Classes](#):

TA	Maciej	Dan	Joie	Christina	Natalia
Time	11:00-12:45	09:00-10:45	09:00-10:45	11:00-12:45	09:00-10:45
Room	HG-12A36	HG-12A36	HG-14A20	NU-4B47	NU-4A67
Group	1	2	3	4	5

- › You have been randomly assigned to one group
 - › Please join the Canvas group that corresponds to your real-life class
 - › Visit a practical together with your submission assignment partner

Prerequisites & Course Materials

> Prerequisites

- > Basic knowledge of statistics and probability theory
- > (Mathematical) Statistics (X_400004 / XB_0049)
Probability Theory (X_400622)
- > Not formal, but useful

> Required readings consist mainly of the [Syllabus](#)

> Class attendance is **not** compulsory, but recommended

> Necessary [software](#)

- > R and RStudio
- > Submission should be .pdf files >> [L^AT_EX](#), [R Markdown](#)
- > Installation guides are on Canvas

Deliverables

Assignments & Quizzes

- There are 7, mostly **biweekly**, assignments
 - An **exception** is Assignment 1, due in **one** week

You will work on the assignments in pairs

- Please join one of the Submission Assignment Groups on Canvas
- Use Discussions to find a teammate if you do not have one yet
- Each assignment consists of a **Quiz** and a **Submission Assignment**

➤ You can work on both with your teammate, however...

-  The **Quiz** has to be submitted **individually** on Canvas
-  The **Submission Assignment** is submitted **to the assignment** on Canvas by **only one of you**
-  Note that the **Quiz** may be randomized to avoid free-riding

Additional Info About Assignments

- › The Quiz will close 15 minutes after deadline (to account for internet issues)
- › There will be a late submission **penalty** for Submission Assignments
 - › The penalty is 2% of Submission Assignment points **per hour**
 - › **Attention:** Canvas rounds the delay **up**
- › If you **cannot** make R Markdown work, you can use a different way to generate a .pdf

Assignment Submission Format

- › For your [Group Submission Assignment](#), upload one .pdf in Canvas
 - › This .pdf should ideally be generated with R Markdown
- › On Canvas/Course Preliminaries, you can find
 - › An R Markdown [Guide File](#) to use as future reference
 - › An R Markdown [Submission](#) Template for the submission assignments
 - › An R Markdown cheat sheet
- › To practice R, there you can also find an [Intro to R](#) file
 - › Unless you have experience with R, you should go through this file before taking the quiz or trying to solve the assignment
 - › However, you do not need to submit anything related to this file
 - › For more comprehensive info, on Canvas you can also find an R manual (namely the first one available from [here](#))

Exams

- › The **Exam** is on Wednesday, 28th of May, at 12:15
- › The **Resit Exam** is on Tuesday, 1st of July, at 08:30
- › Quizzes and assignments help you in preparing
 - › The exam has no programming in **R**
 - › The exam may, however, contain R plots or R output for you to interpret
 - › Interpretation is from the **statistical** point-of-view, not from the programming point-of-view
- › We will devote part of the last class before the exam as revision
- › The **best** way to prepare for the exam is to study the Syllabus
- › You do not need to learn all formulas by heart
 - › You will be provided with formulas for more 'obscure' quantities when necessary
 - › But you are supposed to know **what** a certain quantity is/does
 - › E.g. how do you interpret a median, what is it used for, . . .

Final Grade

- › The **final grade** is given by $0.6 \cdot E + 0.4 \cdot A$ rounded to the closest half-integer
 - › However, the **condition** to pass the course (when considering the final exam) is $\min(A, E) \geq 5.5$
 - › If that is **not satisfied**, the final grade is given by $\min(A, E)$
- › E is the **exam grade**, one of the following:
 - › The final exam grade
 - › The resit exam grade
- › A is an average **assignment grade** computed as $\max\left(1, \frac{\sum_{i=1}^7 A_i}{13}\right)$
 - › $A_i = Q_i + S_i$ is the score (in points) of assignment i , $i = 1, \dots, 7$
 - › Assignment 1 is worth a maximum of 10 points
 - › Assignments 2-7 are worth a maximum of 20 points each

Main Lecture Schedule & Assignment Deadlines

Date	Topic	Assignment	Assignment Deadline
Wed, Feb 05, 2025	Organisation Ch. 1 & 2 of Syllabus	A1 is released	Tue, Feb 11, 2025 at 23:59
Wed, Feb 12, 2025	Discussion of A1 Ch. 2 & 3 of Syllabus	A2 is released	Tue, Feb 25, 2025 at 23:59
Wed, Feb 19, 2025	Ch. 3 & 4 of Syllabus	⤴	⤴
Wed, Feb 26, 2025	Discussion of A2 Ch. 4 of Syllabus	A3 is released	Tue, Mar 11, 2025 at 23:59
Wed, Mar 05, 2025	Ch. 5 of Syllabus	⤴	⤴
Wed, Mar 12, 2025	Discussion of A3 Ch. 5 of Syllabus	A4 is released	Tue, Apr 01, 2025 at 23:59
Wed, Mar 19, 2025	Ch. 6 of Syllabus Exam Preparation (for Ch. 1-5)	- - -	- - -
Wed, Apr 02, 2025	Discussion of A4 Ch. 6 of Syllabus	A5 is released	Tue, Apr 15, 2025 at 23:59
Wed, Apr 09, 2025	Ch. 6 & 7 of Syllabus	⤴	⤴
Wed, Apr 16, 2025	Discussion of A5 Ch. 7 & 8 of Syllabus	A6 is released	Tue, May 06, 2025 at 23:59
Wed, Apr 23, 2025	Ch. 8 of Syllabus	⤴	⤴
Wed, May 07, 2025	Discussion of A6 Ch. 8 of Syllabus	A7 is released	Tue, May 20, 2025 at 23:59
Wed, May 14, 2025	Ch. 8 of Syllabus	⤴	⤴
Wed, May 21, 2025	Discussion of A7 Exam Preparation (for Ch. 6-8)	⤴	⤴
Wed, May 28, 2025	Final Exam	- - -	- - -
Tue, Jul 01, 2025	Resit Exam	- - -	- - -

Help Needed? 

- › For general questions, using the [Discussions](#) on Canvas is very efficient
- › For organisational matters, contact me at [✉ m.a.corstanje@vu.nl](mailto:m.a.corstanje@vu.nl)
- › There are no fixed office hours
 - › You are always welcome to ask questions during/following the lecture
 - › A meeting can be requested in extraordinary cases
- › Make use of the practicals to ask questions about the assignments
 - › Make sure to come [prepared](#) to the practicals
 - › The practicals are there to practice R itself, not R Markdown: for that, you can use the related [Discussion](#) on Canvas